

1 Solution — GIAM 8.3

Problem 6

1.1 Problem

Verify that the final deduction in the proof of Cantor's theorem,

$$(y \in S \implies y \notin S) \wedge (y \notin S \implies y \in S),$$

is truly a contradiction.

1.2 Setup — What “Contradiction” Means

In propositional logic, a *contradiction* is a statement whose truth value is *False* under every assignment of truth values to its atomic propositions. (Equivalently, the statement is logically equivalent to $\perp$, and its negation is a *tautology*.)

The statement in the problem involves a single atomic proposition: $P \equiv "y \in S."$ Then $y \notin S$ is just $\neg P$, and the statement becomes

$$\Phi := (P \implies \neg P) \wedge (\neg P \implies P).$$

We must show Φ is False for every truth assignment to P . Since P takes two values (T or F), there are just two cases to check.

1.3 Approach 1 — Truth Table

The complete truth table:

Truth Table for $(P \Rightarrow \neg P) \wedge (\neg P \Rightarrow P)$

Letting P abbreviate the statement " $y \in S$ "

P	$\neg P$	$P \Rightarrow \neg P$	$\neg P \Rightarrow P$	$(P \Rightarrow \neg P) \wedge (\neg P \Rightarrow P)$
T	F	F	T	F
F	T	T	F	F

Final column is F in every row.

A statement that's False under every truth assignment is, by definition, a contradiction.

$$(P \Rightarrow \neg P) \wedge (\neg P \Rightarrow P) \equiv \perp \text{ (contradiction)}$$

Figure 1.1: Truth table for the statement $\Phi = (P \Rightarrow \neg P) \wedge (\neg P \Rightarrow P)$, where P abbreviates ' $y \in S$ '. The final column is F under both truth assignments to P , so Φ is False everywhere — it is a contradiction.

Reading off:

P	$\neg P$	$P \Rightarrow \neg P$	$\neg P \Rightarrow P$	Φ
T	F	F	T	F
F	T	T	F	F

Table 1.1

The final column is False in every row, so Φ is False under every truth assignment. By definition, Φ is a contradiction. ■

(Quick reminder on implications: $A \Rightarrow B$ is False only when A is True and B is False; otherwise it's True. So $P \Rightarrow \neg P$ is False exactly when P is True and $\neg P$ is False — i.e., P is True. And $\neg P \Rightarrow P$ is False exactly when $\neg P$ is True and P is False — i.e., P is False.)

1.4 Approach 2 — Direct Case Analysis

A purely deductive argument:

Case 1: Suppose P . From $P \Rightarrow \neg P$ (the left conjunct) and P , modus ponens gives $\neg P$. So we have both P and $\neg P$, the explicit contradiction $P \wedge \neg P$.

Case 2: Suppose $\neg P$. From $\neg P \Rightarrow P$ (the right conjunct) and $\neg P$, modus ponens gives P . So we have both $\neg P$ and P , the explicit contradiction $P \wedge \neg P$.

Either P or $\neg P$ holds (law of excluded middle), and either way we derive $P \wedge \neg P$. So Φ entails the canonical contradiction $P \wedge \neg P$, hence is unsatisfiable. ■

1.5 Approach 3 — Equivalence to $P \Leftrightarrow \neg P$

The conjunction of $A \Rightarrow B$ and $B \Rightarrow A$ is just the biconditional $A \Leftrightarrow B$. Applied to our statement with $A = P$ and $B = \neg P$:

$$\Phi = (P \Rightarrow \neg P) \wedge (\neg P \Rightarrow P) \equiv (P \Leftrightarrow \neg P).$$

A biconditional $A \Leftrightarrow B$ is True iff A and B have the *same* truth value. But P and $\neg P$ have *opposite* truth values by the very definition of negation — so $P \Leftrightarrow \neg P$ is False in every truth assignment.

So $\Phi \equiv (P \Leftrightarrow \neg P) \equiv \perp$. ■

This third view is arguably the most illuminating: the “final deduction” in Cantor’s proof asserts that $y \in S$ holds if and only if $y \in S$ doesn’t hold. That’s $P \Leftrightarrow \neg P$ — a textbook impossibility.

1.6 Why Cantor’s Proof Arrives at This Specific Form

Recap how the formula arises in Theorem 8.3.1. Suppose $f : A \rightarrow \mathcal{P}(A)$ is given and define $S = \{x \in A : x \notin f(x)\}$. Suppose toward contradiction that $S = f(y)$ for some $y \in A$. Then:

- By the *definition* of S : $y \in S \Leftrightarrow y \notin f(y)$.
- By the *hypothesis* $S = f(y)$: $y \in f(y) \Leftrightarrow y \in S$.

Chaining these biconditionals: $y \in S \Leftrightarrow y \notin f(y) \Leftrightarrow y \notin S$.

The textbook writes the same chain as a pair of *implications* (slightly more explicit for the reader):

$$\begin{aligned} y \in S &\implies y \in f(y) \implies y \notin S, \\ y \notin S &\implies y \notin f(y) \implies y \in S. \end{aligned}$$

The conjunction of these two derived implications is Φ — exactly the statement we verified is a contradiction. Hence the hypothesis $S = f(y)$ leads to $\perp$, so no such y exists, so $S \notin \text{image}(f)$, so f is not surjective. This is the heart of Cantor's theorem.

1.7 Remark — Why Bother Writing Both Implications?

Logically, the two implications $P \Rightarrow \neg P$ and $\neg P \Rightarrow P$ are jointly equivalent to the single biconditional $P \Leftrightarrow \neg P$, which is *visibly* contradictory. So why does the textbook write both directions?

It's a pedagogical choice. Showing the *two* implications mirrors the two halves of Cantor's argument:

- *Left half* (the definition of S forces something out): if $y \in S$ then by hypothesis $y \in f(y)$, but the *defining condition* of S means $y \notin f(y)$, so $y \notin S$.
- *Right half* (the definition forces something in): if $y \notin S$ then by hypothesis $y \notin f(y)$, but the *defining condition* means $y \in f(y)$, so $y \in S$.

Each half is constructed differently — one uses the definition to *exclude* y from S , the other to *include* it — and writing both explicitly makes the symmetry of the diagonal trick visible. The collapse into " $P \Leftrightarrow \neg P$ " happens at the very last step.

1.8 Remark — A Shorter Argument from One Implication

Curiously, *just one* of the two implications already suffices in classical propositional logic. From $P \Rightarrow \neg P$ alone:

- Suppose P . Then $\neg P$ by the implication. So $P \wedge \neg P$. Contradiction.
- Hence (by reductio) $\neg P$.

But this gives only $\neg P$, not a contradiction with the listing assumption. The other implication $\neg P \Rightarrow P$ is needed to deduce P as well, producing the full contradiction $P \wedge$

$\neg P$. So both directions are *individually* “half” of the contradiction; together they pin it down completely.

(In intuitionistic logic — where reductio for $\neg P$ but not for P is allowed — the symmetry is more delicate. Cantor’s argument as stated remains valid because both implications are used constructively to derive both halves of the contradiction.)

1.9 Remark — A Tautology in Disguise

The negation of Φ is the tautology

$$\neg\Phi \equiv \neg((P \Rightarrow \neg P) \wedge (\neg P \Rightarrow P)) \equiv (P \wedge P) \vee (\neg P \wedge \neg P) \equiv P \vee \neg P,$$

which is the *law of excluded middle*. This is satisfying: the fact that Φ is unsatisfiable is *equivalent* to the assertion that every proposition is either true or false — the most fundamental two-valued principle of classical logic. Cantor’s argument is, at root, classical logic doing its most basic job.

So Problem 6 is really verifying that *classical bivalence* and Φ are incompatible — which is a fancy way of saying Φ has no models. Either statement of the fact, truth-table or excluded-middle, confirms that the final deduction in Cantor’s proof is indeed a *bona fide* contradiction.